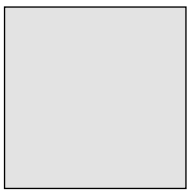
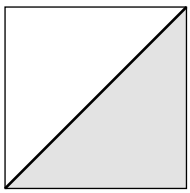
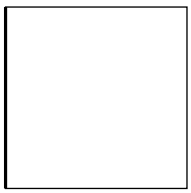
Use whole squares inside the gray shape. Cover all squares with some gray area. A touch at an edge or corner adds no square.



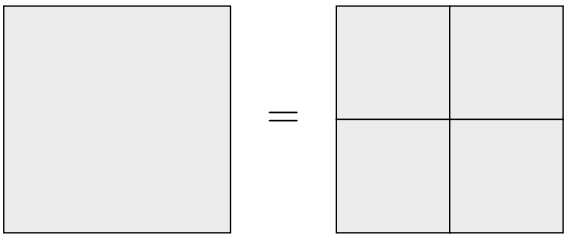
whole



partial

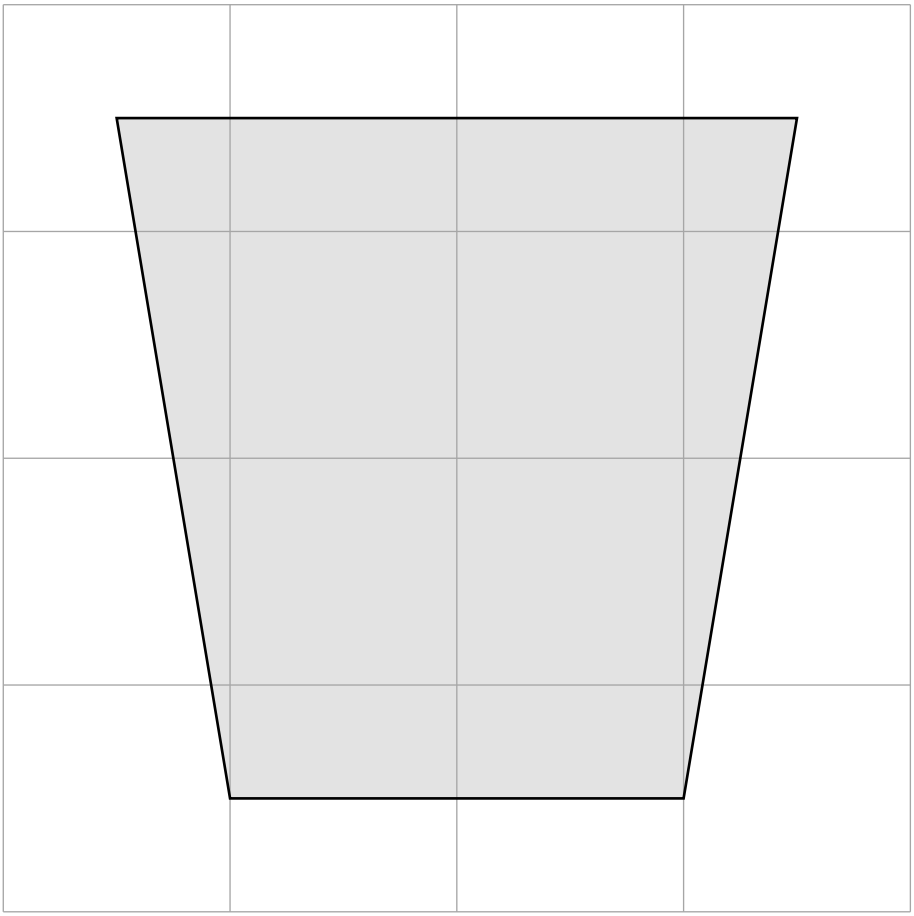


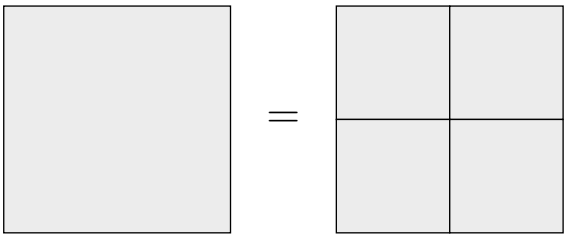
outside



1 big square
= 4 small squares

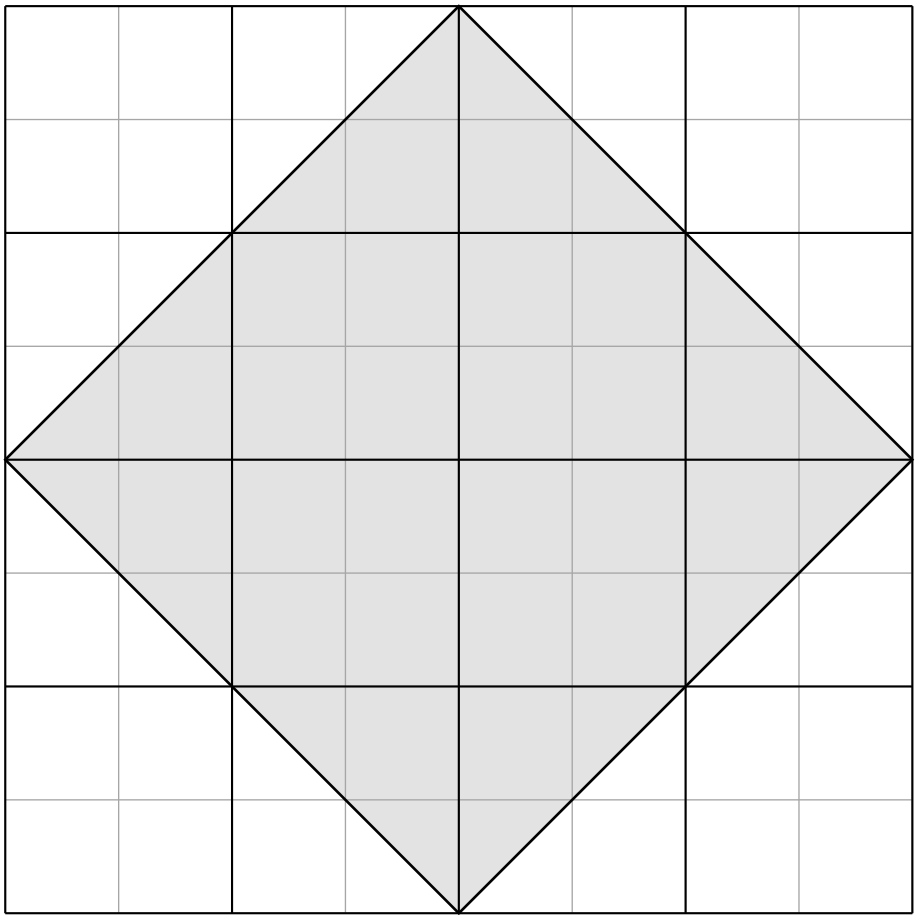
Problem 2: Cover this shape, then split two big tiles into four small tiles each. Choose the two big tiles that let you remove the most small tiles while keeping the gray shape covered.



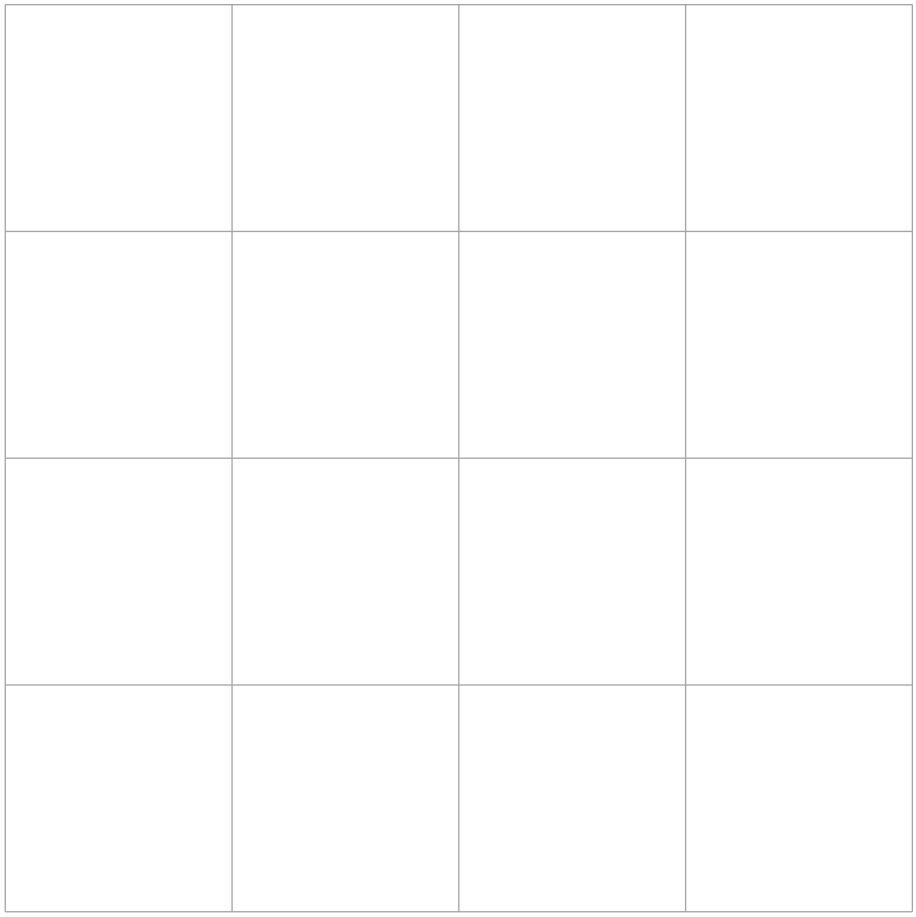


1 big square
= 4 small squares

Problem 4: Change the diamond so four outside small squares become partly gray. Keep every big square whole, partial, or outside as it was.



Problem 5: Draw two shapes, one at a time, each with 3 whole and 3 partly gray squares. Make one with less than 4 big squares of gray and one with more than 5.



Problem 6: Keep the same whole and partial squares. Can you make the larger shape bigger again without filling another whole square?

One grid square is 1 big-square unit.